Xiao **Zhang**

(+1) 737-382-2777 | zx123@utexas.edu | timez-zx.github.io | Timez-zx

About me

I am a Ph.D. student at UT Austin advised by Prof. Daehyeok Kim, working at the intersection of networked/distributed systems and AI infrastructure. My work spans datacenter networks, 5G and edge computing, and cross-layer telemetry, with a current focus on making AI workloads predictable and efficient—from the network fabric up to GPU resources. I build practical, measurement-driven systems that bridge real-world deployment with the infrastructure needs of modern AI.

Experience

Google Cloud — *Sunnyvale, CA*
Software Engineering Intern — *June 2026 - Present*

- Developing network telemetry for LLM workload analysis

University of Texas at Austin (with Prof. Daehyeok Kim) — *Austin, TX*
Ph.D. Student — *Sept. 2024 - Present*

- Designed and built a private 5G testbed with GPU-accelerated edge servers for studying AI inference latency over cellular networks
- Conducted large-scale measurements using AWS Wavelength Zones and Verizon 5G to identify latency variability in edge AI applications
- Identified key performance bottlenecks in wireless scheduling and GPU resource contention; currently developing cross-layer resource management techniques for predictable edge inference latency

University of Pennsylvania (with Prof. Vincent Liu) — *Philadelphia, USA*
Visiting Student — *July. 2023 - Present*

- *Beaver*: Enabling Practical Distributed Snapshots Exploiting Software Load Balancers

Shanghai Jiao Tong University (with Prof. Shizhen Zhao) — *Shanghai, China*
Master Student — *Sep. 2021 - July. 2023*

- *Flattened Clos Plus (FC+)*: Near-optimal topology-routing co-design free of deadlocks for RoCE-based expander networks
- *Flattened Clos (FC)*: Deadlock-free topology-routing co-design for RoCE-based expander networks

Honors and Awards

2025	**Amazon AI Fellowship**, Awardee	*Austin, TX*
2021	**Outstanding Graduate of Shanghai**, Awardee	*Shanghai*
2020	**Liu Yongling Scholarship**, Awardee	*Shanghai*

Education

University of Texas at Austin — *Austin, Texas*
Ph.D. in Computer Science — *Sept. 2024 - Present*

- Advisor: Prof. Daehyeok Kim
- Research focus: Networked & distributed systems, AI infrastructure, Edge computing

Shanghai Jiao Tong University *Shanghai, China*
M.E. in Communication Engineering *Sept. 2021 - May. 2024*

- Thesis title: FC+: Near-optimal Deadlock-free Expander Data Center Networks

Shanghai Jiao Tong University *Shanghai, China*
B.E. in Information Engineering *Sept. 2017 - June. 2021*

- Thesis title: Design of Robust and Efficient Edge Server Placement and Server Scheduling Policies

Publication

- **Xiao Zhang**, Daehyeok Kim. Enabling SLO-Aware 5G Multi-Access Edge Computing with SMEC. In Proceedings of 23rd USENIX Symposium on Networked Systems Design and Implementation **(NSDI)**, May 2026.
- Liangcheng Yu, **Xiao Zhang**, Haoran Zhang, John Sonchack, Dan Ports, Vincent Liu. Beaver: Practical Partial Snapshots for Distributed Cloud Services. In Proceedings of 18th USENIX Symposium on Operating Systems Design and Implementation **(OSDI)**, July 2024.
- **Xiao Zhang**, Peirui Cao, Yongxi Lyu, Qizhou Zhang, Shizhen Zhao, Xinbing Wang, Chenghu Zhou. FC+: Near-optimal Deadlock-free Expander Data Center Networks. In Proceedings of 21st IEEE International Symposium on Parallel and Distributed Processing with Applications **(ISPA)**, December 2023.
- Shizhen Zhao*, Qizhou Zhang*, Peirui Cao, **Xiao Zhang**, Xinbing Wang, Chenghu Zhou. Flattened Clos: Designing High-performance Deadlock-free Expander Data Center Networks Using Graph Contraction. In Proceedings of 20th USENIX Symposium on Networked Systems Design and Implementation **(NSDI)**, 2023.
- Shizhen Zhao*, **Xiao Zhang***, Peirui Cao, Xinbing Wang. Design of Robust and Efficient Edge Server Placement and Server Scheduling Policies. In Proceedings of IEEE/ACM 29th International Workshop on Quality of Service **(IWQoS)**, 2021.

Skills

Programming	C/C++, Python, Verilog, VHDL, Matlab
Tools	eBPF, DPDK, FPGA, Network Simulator 3 (NS-3), AWS
Languages	English, Chinese